

CSIS 4495 – Applied Research Project

Progress Report 5

GREENPATH – Sustainable Products Website with AI Chatbot Integration

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Work Date/Hours Log

The table below shows summary of the work done till date with the respective time spent on each task.

Date	Number of Hours	Description of Work Done
2025-11-08	1.5	Implemented Chat UI redesign by adding bot icon, refactored logic.
2025-11-10	1.5	Implemented Chat UI redesign by stabilized scroll container behavior, improved markdown formatting and message parsing pipeline.
2025-11-11	1	Worked on my project video presentation.
2025-11-12	1	Worked on my project video presentation (contd).
2025-11-14	1.5	Revalidated ML logic including Holt Linear Trend forecasting, drift-based prediction.
2025-11-17	1.5	Worked on the Z-score anomaly detection, and k-means clustering on normalized sector shares.
2025-11-18	1.5	Improved Chat assistant dataset alignment: refined prompt engineering for agg.csv emissions queries.
2025-11-19	1.5	Tested emission-specific response generation and completed final UI consistency checks.

Description of Work Done

In the past weeks, I focused on strengthening GreenPath's technical foundation by improving UI stability, restoring critical analytics logic, and deepening machine learning functionality across the platform. Major work included enhancing the Chat interface with a bot avatar, improved bubble rendering, markdown cleanup, and stabilized scroll management to isolate scroll state from the page layout. Significant technical effort was invested into recovering and validating Analytics.tsx and AnalyticsML.tsx after accidental overwrite; this involved reconstructing Holt Linear Trend forecasting logic, reimplementing drift-based next-year prediction, recalibrating anomaly detection through Z-score calculations on year over year deltas and restoring k-means clustering on normalized sectoral compositions. Additional improvements were applied across the Home and Categories pages, including layout restructuring, larger carousel rendering, responsive media handling, and background layering tests using CSS stacking context and container boundaries. I also enhanced Chat assistant accuracy by refining dataset aligned prompts and verifying emissions-based responses against the agg.csv dataset. A midterm video presentation was created by recording multiple segments demonstrating analytics, frontend behavior, and ML outputs, which were combined into a single final submission. Challenges included layout conflicts preventing Category page background rendering, and early chat scrolling issues, all of which were fixed through targeted technical adjustments to file structure, layout containers, and component logic. Overall, this phase involved substantial technical work across data processing, ML model behavior, UI engineering, and system reliability

Repo Check in of Implementation Completed

The files/folders I have checked in the repo since report 2 are as follows:

Recent committed files under Implementation/greenpath/ include:

src/pages/Chat.tsx – Refined chat layout with bot icon, stabilized scroll container, improved Markdown rendering, and enhanced dataset-aligned responses.

src/pages/AnalyticsML.tsx – Added explanatory sections for ML modules, improved layout, and refined forecasting, anomaly detection, prediction, and clustering outputs.

src/pages/Home.tsx – Expanded carousel visuals, improved hero section responsiveness, and updated page content for clearer project introduction.

src/pages/Categories.tsx – Attempted background enhancements and refined UI elements for better readability and structure.

src/layouts/RootLayout.tsx – Updated global layout styling and ensured consistency with new UI adjustments across pages.

src/router.tsx – Minor routing refinements supporting improved page transitions and layout behavior.